

EXPERIMENT NO. 5

AIM

Determination of the equivalent circuit parameters of a single phase transformer.

OBJECTIVE

- To obtain the equivalent circuit parameters from Open Circuit and Short Circuit tests.
- To calculate the efficiency as well as regulation of the transformer at given load condition.
- To conduct Sumpner's test on two identical single phase transformer & to verify the results with OC & SC tests.
- To obtain the B-H curve for a single-phase transformer.

THEORY

The physical basis of the transformer is mutual induction between two circuits linked by a common magnetic field. Transformer is required to pass electrical energy from one circuit to another, via the medium of the pulsating magnetic field, efficiently and economically as possible. This could be achieved using either iron or steel which serves as a good permeable path for the mutual magnetic flux.

The principle of operation of this circuit can be explained as follows:

Let an alternating voltage v_1 be applied to a primary coil of N_1 turns linking a suitable iron core. A current flows in the coil, establishing a flux Φ_p in the core.

This flux induces an emf e_1 in the coil to counterbalance the applied voltage v_1 .
This e.m.f. is

{1}

Assuming sinusoidal time variation of the flux, let $\Phi_p = \Phi_m \sin \omega t$. Then,

$$e_1 = N_1 \omega \Phi_m \cos \omega t \quad \{2\}$$

The r.m.s. value of this voltage is given by:

$$E_1 = 4.44 f N_1 \Phi_{m1} \quad \{3\}$$

Now if there is a secondary coil of N_2 turns, wound on the same core, then by mutual induction an emf e_2 is developed therein. The r.m.s. value of this voltage is given by:

$$E_2 = 4.44 f N_2 \Phi_{m2} \quad \{4\}$$

Where Φ_{m2} is the maximum value of the (sinusoidal) flux linking the secondary coil (Φ_s). If it is assumed that $\Phi_p = \Phi_s$ then the primary and secondary e.m.f.'s bear the following ratio:

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} \quad \{5\}$$

But, in actual practice, $\Phi_p \neq \Phi_s$ since some of the flux paths linking the primary coil do not link the secondary coil and similarly some of the flux paths linking

the secondary coil do not link the primary coil. The fluxes which do not link both the coils are called the leakage fluxes of the primary and secondary coil.

In a practical transformer a very large proportion of the primary and secondary flux paths are common and leakage fluxes are comparatively small. Therefore $\Phi_p \approx \Phi_s$.

If in addition, winding resistances are neglected being usually small in a practical transformer, then

$$V_1 \approx E_1 \quad \{6\}$$

$$V_2 \approx E_2 \quad \{7\}$$

Although the iron core is highly permeable, it is not possible to generate a magnetic field in it without the application of a small m.m.f (magneto-motive force), denoted by M_m . Thus even when the secondary winding is open circuited, a small magnetizing current (i_m) is needed to maintain the magnetic flux. The current of the primary circuit on no-load is of the order of 5% of full load current. Also, the pulsation of flux in the core is productive of core loss, due to hysteresis and eddy currents. These losses are given by:

$$P_h = K_h B_{\max}^{1.6} f \quad \{8\}$$

$$P_e = K_e B_{\max}^2 f^2 \quad \{9\}$$

$$P_c = P_e + P_h \quad \{10\}$$

Where P_h , P_e and P_c are hysteresis, eddy current and core losses respectively, K_h and K_e are constants which depend on the magnetic material, and $B_{\max}$ is the

maximum flux density in the core. These losses will remain almost constant if the supply voltage and frequency are held constant. The continuous loss of energy in the core requires a continuous supply from the electrical source to which the primary is connected. Therefore, there must be a current component i_c which accounts for these losses. It should be noted that magnetizing current (i_m) and core loss component of current (i_c) are perpendicular to each other. The resultant of these two currents is the no-load current i_0 . Generally the magnitude of this current is very small compared to that of the rated current of the transformer (of the order of 5% of the rated current). This current makes a phase angle Φ_0 of the order of $(\cos^{-1}(0.2))$ with the applied voltage.

If a load of finite impedance is connected across the second coil, a current i_2 will flow through it. This tends to alter the mmf and thereby the flux in the core. But this is prevented by an immediate and automatic adjustment of the primary current i_1 , thereby maintaining the flux Φ at the original value. This value of flux is required to produce the emf of self-induction e_1 . Any reduction of the flux would cause a reduction of e_1 , leaving a voltage difference between v_1 and e_1 which would be sufficient to increase the primary current and thereby re-establish the flux. Thus any current which flows in the secondary causes its counterpart to flow in the primary so that the flux Φ (and therefore the mmf $-M_m$) shall always be maintained at a value such that the voltage applied v_1 to the primary terminals shall be balanced by the induced emf e_1 (neglecting voltage drops due to resistance and leakage flux effects). Thus if

current flows in the secondary (i_2), then $i_1 = i_0 + \frac{N_2}{N_1} i_2$ so that effective mmf in the core remains at M_m .

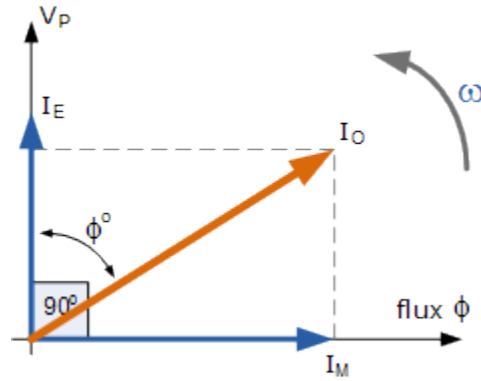


Figure 1: No Load Phasor Diagram

In phasor notation:

$$i_1 = i_0 + \frac{N_2}{N_1} i_2 \quad \{11\}$$

i_0 is quite small compared to the rated current and is usually neglected if transformer is loaded. Thus:

$$i_1 \cong \frac{N_2}{N_1} i_2 \quad \{12\}$$

Thus by making the assumptions:

- Winding resistances are small
- Magnetizing current is small
- Core losses are small
- Leakage fluxes are small

We can write:

$$\frac{I_1}{I_2} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{V_2}{V_1} \quad \{13\}$$

EQUIVALENT CIRCUIT OF A PRACTICAL TRANSFORMER

The practical transformer has coils of finite resistance. Though this resistance is distributed uniformly, it can be taken as concentrated. Also, all the flux produced by the primary current cannot be confined into a desired path completely as an electric current. Though a greater proportion links both the coils (known as mutual flux), a small proportion called the leakage flux links one or other winding, but not both. It does not contribute to the transfer of energy from primary to secondary. On account of the leakage flux, both the windings have a voltage drop which is due to leakage reactance. The transformer shown in Figure 1 can be resolved into an equivalent circuit as shown in Figure 2 in which the resistance and leakage reactance of primary and secondary respectively are represented by lumped R_1 , X_1 , R_2 and X_2 .

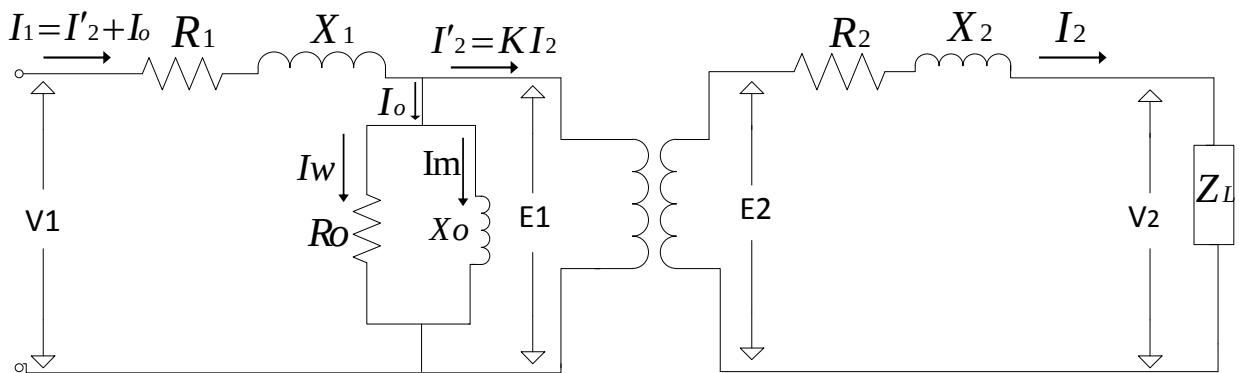


Figure 2: Equivalent Circuit

This equivalent circuit can be further simplified by referring all quantities in the secondary side of the transformer to primary side and is shown in Figure 3.

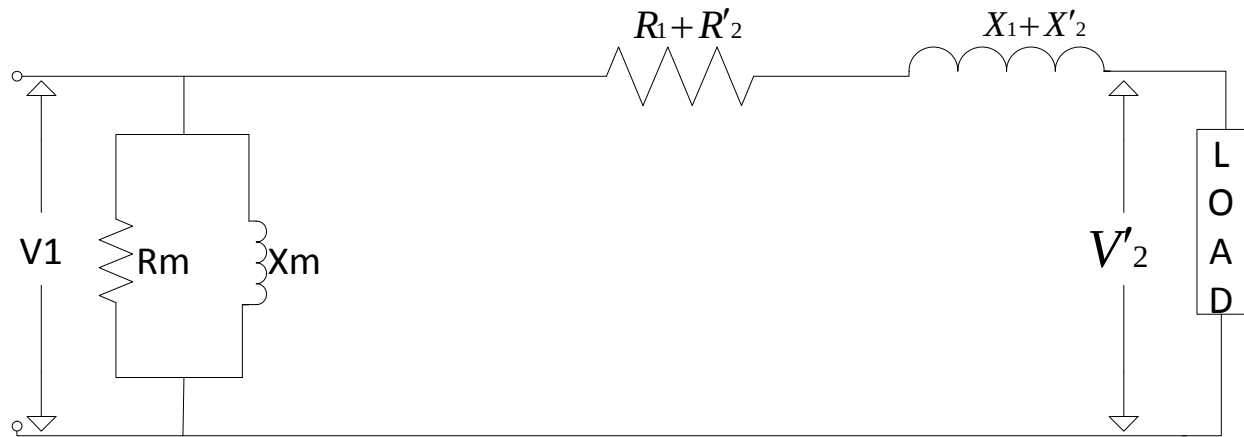


Figure 3: Equivalent Circuit

DETERMINATION OF EQUIVALENT CIRCUIT PARAMETERS

The equivalent circuit parameters can be determined by performing tests that involve little power consumption. Two tests, a no-load test (or open circuit test) and short circuit test will provide data for determining the parameters of the equivalent circuit.

SHORT CIRCUIT TEST

Consider the circuit shown in Figure 3. Suppose the input voltage is reduced to a small fraction of rated value and secondary terminals are short-circuited. A current will circulate in the secondary winding. Since a small fraction of

rated voltage is applied to the primary winding, the flux in the core and hence the core loss is very small. Hence, the power input on short circuit is dissipated as heat in the winding. The circuit diagram is shown in Figure 4.

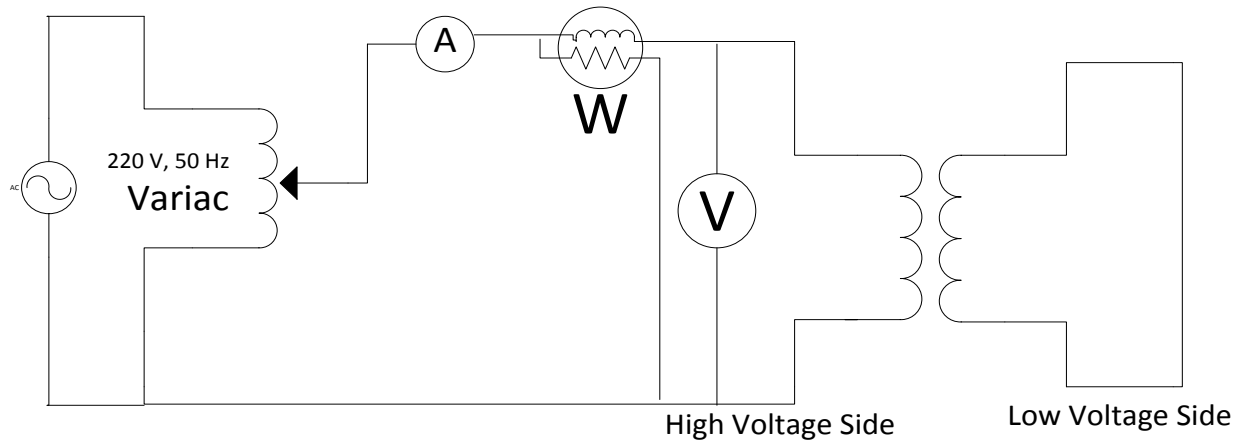


Figure 4: Short circuit test diagram

In this test, the LV terminals of the transformer are short circuited. The primary voltage is gradually applied till the rated current flows in the winding. Since, the applied voltage is very small (may be of the order of 5-8%), the magnetizing branch can now be eliminated from the equivalent circuit. The modified equivalent circuit is shown in Figure 5.

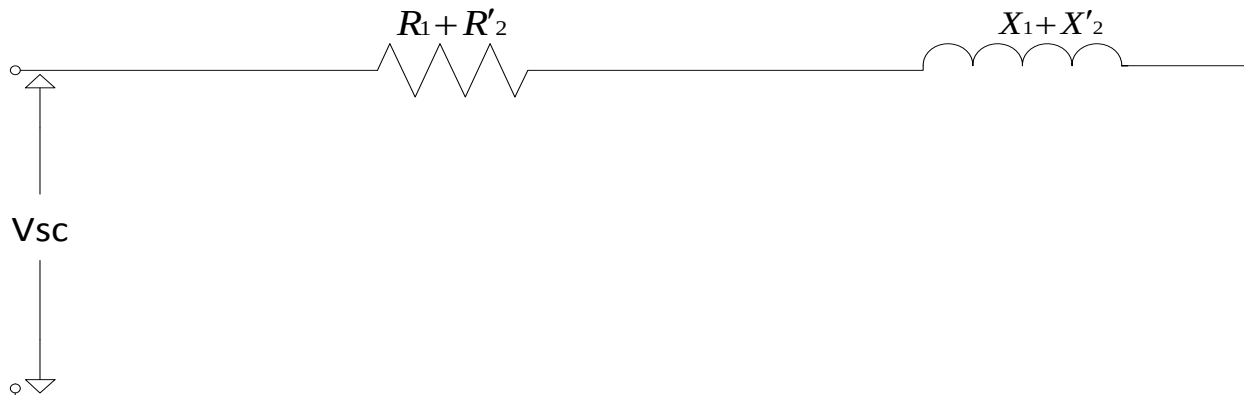


Figure 5: Equivalent circuit diagram of short circuit test

If V_{sc} is the applied voltage to circulate the rated current (I_2') on short circuit, and P_c is the power input to the transformer then,

$$Z_{sc} = \frac{V_{sc}}{I_2'} \quad \{14\}$$

$$\cos \theta = \frac{P_c}{V_{sc} I_2'} \quad \{15\}$$

Therefore,

$$R_1 + R_2' = Z_{sc} \cos \theta \quad \{16\}$$

$$X_1 + X_2' = Z_{sc} \sin \theta \quad \{17\}$$

PROCEDURE

- Set up the circuit as shown in figure. Here, the secondary terminals are shorted. Use appropriate range for current coil of wattmeter, 5 Ampere range for ammeter and 150 Volts range for the potential coil of the wattmeter. Only a low voltage is to be applied to the primary of the transformer. This is done by setting variac output to zero initially and increasing it very slowly. If normal voltage is applied to the transformer under short circuit condition, a very large current will flow through it causing damage to the windings, the ammeter and the wattmeter. Using a multimeter, a small voltage (maximum of 24 Volts) is to be supplied to the transformer for the test.
- Measure the voltage from the multimeter reading. Adjust the variac to obtain full load current through the transformer primary

OPEN CIRCUIT/NO LOAD TEST

The shunt branch parameters are determined by performing this test. Since, the core loss and the magnetizing current depend on applied voltage, this test is performed by applying the rated voltage to one of the windings keeping the other winding open (generally HV winding is kept open and rated voltage is applied to LV winding). The circuit diagram to conduct this test is shown in Figure 6.

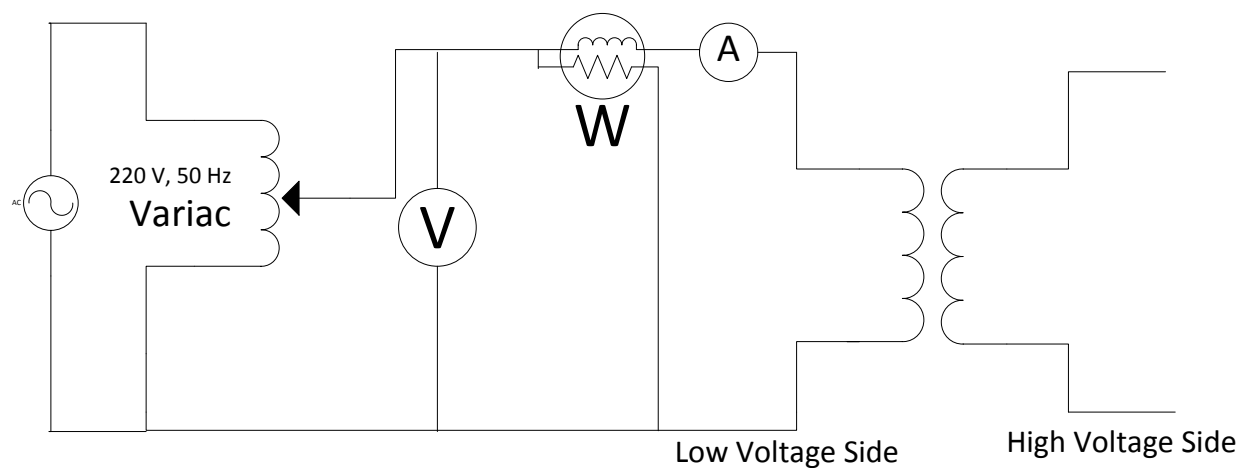


Figure 6: Open circuit test diagram

Since, the secondary terminals are open (no load is connected across the secondary), the current drawn from the source is called as no load current. On no-load, the approximate equivalent circuit shown in Figure 6 can be further reduced and is shown in Figure 7.

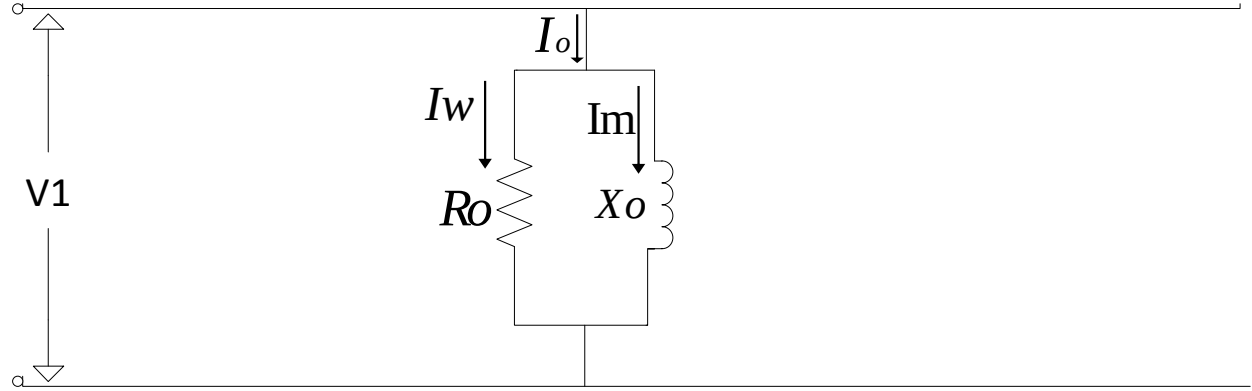


Figure 7: Equivalent circuit diagram of open circuit test

Under no-load condition the power input to the transformer is equal to the sum of losses in the primary winding resistance R_1 , and core loss. Since, no-load current is very small, the loss in winding resistance is neglected. Hence, on no-load the power drawn from the source is dissipated as heat in the core. If I_o and P_i are the current and input power drawn by the transformer at rated voltage V_1 respectively, then

$$\cos \phi_o = \frac{P_i}{V_1 i_o} \quad \{18\}$$

$$i_c = i_o \cos \phi_o \quad \{19\}$$

$$i_m = i_o \sin \phi_o \quad \{20\}$$

Therefore,

$$R_o = \frac{V_1}{i_c} \quad \{21\}$$

$$X_m = \frac{V_1}{i_m} \quad \{22\}$$

PROCEDURE

- Set up the circuit as shown in the figure. The secondary terminals are left open. Use 1 Ampere range of ammeter and the current coil of the wattmeter, and 300 Volts range for the potential coil of the wattmeter.
- Adjust the voltage to the transformer to 240 Volts using the variac and the multimeter. Note down the meter readings.

SUMPNER'S TEST

The Sumpner's test is another method of determining efficiency, regulation and heating under load conditions. The O.C. and S.C. tests give us the equivalent circuit parameters but cannot give heating information under various load conditions. The Sumpner's test gives heating information also. In O.C. test, there is no load on the transformer while in S.C. circuit test also only fractional load gets applied. In all in O.C. and S.C. tests, the loading conditions are absent. Hence the results are inaccurate. In Sumpner's test, actual loading conditions are simulated hence the results obtained are much more accurate. Thus Sumpner's test is much improved method of predetermining regulation and efficiency than O.C. and S.C. tests.

The Sumpner's test requires two identical transformers. Both the transformers are connected to the supply such that one transformer is loaded on the other. Thus power taken from the supply is that much necessary for supplying the losses of both the transformers and there is very small loss in the control circuit.

While conducting this test, the primaries of the two identical transformers are connected in parallel across the supply V_1 . While the secondaries are connected in series opposition so that induced e.m.f.s in the two secondaries oppose each other. The secondaries are supplied from another low voltage supply are connected in each circuit to get the readings. The connection diagram is shown in the Figure 8.

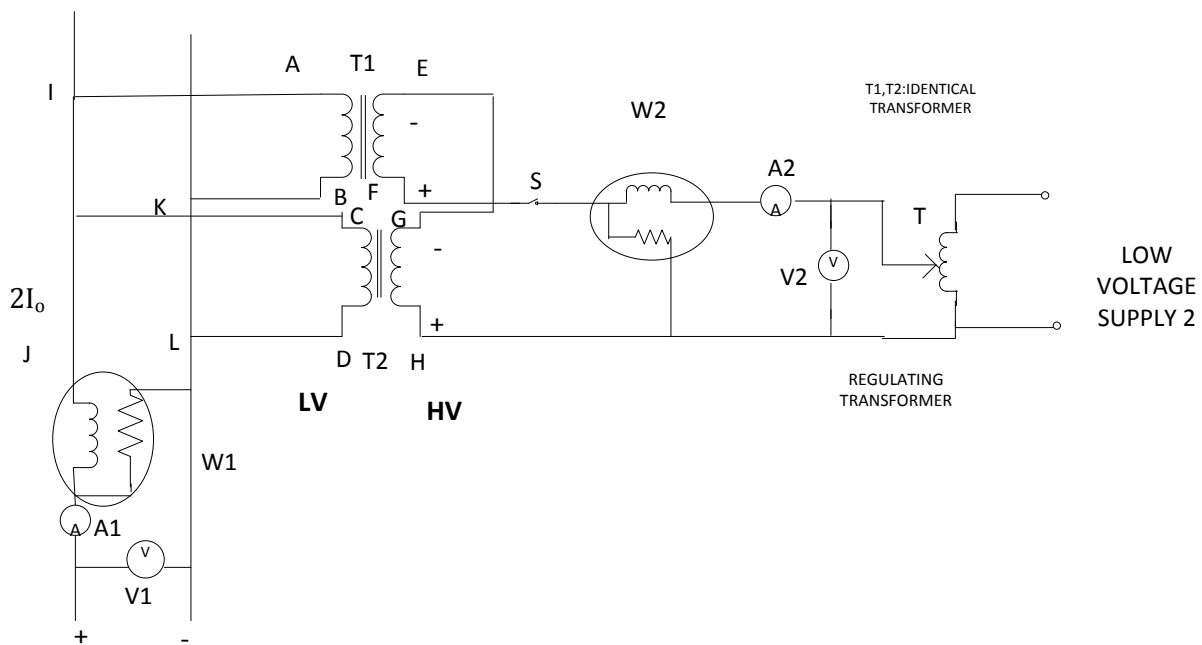


Figure 8: Circuit Diagram for Sumpner test

T_1 and T_2 are two identical transformers. The secondaries of T_1 and T_2 are connected in series opposition. So $E_{EF} = E_{GH}$ i.e. induced in two secondaries are equal but the secondaries are connected such that E is connected to G and

F is connected to H. Due to such series opposition, two e.m.f.s act in opposite direction to each other and cancel each other. So net voltage on the local circuit of secondaries is zero, when primaries are excited by supply 1 of rated voltage and frequency. So there is no current flowing in the loop formed by two secondaries. The series opposition can be checked by another voltmeter connected in the secondary circuit as per polarity test. If it reads zero, the secondaries are in series opposition and if it reads double the induced e.m.f. in each secondary, it is necessary to reverse the connections of one of the secondaries.

As per superposition theorem, if V_2 is assumed zero then due to phase opposition to current flows through secondary and both the transformers T_1 , T_2 are as good as on no load. So O.C. test gets simulated. The current drawn from source V_1 in such case is $2I_0$ where I_0 is no load current of each transformer. The input power as measured by wattmeter W_1 thus reads the iron losses of both the transformers.

$$\therefore P_i \text{ per transformer} = W_1 / 2 \quad \text{as } T_1, T_2 \text{ are identical}$$

Then a small voltage V_2 is injected into the secondary with the help of low voltage transformer, by closing the switch S. With regulation mechanism, the voltage V_2 is adjusted so that the rated secondary current I_2 flows through the secondaries as shown. I_2 flows from E to F and then from H to G. The flow of I_1 is restricted to the loop B A I J C D L K B and it does not pass through W_1 . Hence W_1 continues to read core losses. Both primaries and secondaries carry rated current so S.C. test condition gets simulated. Thus the wattmeter W_2 reads the total full load copper losses of both the transformers.

$$\therefore (P_{cu}) \text{ F.L. per transformer} = W_2 / 2$$

PROCEDURE

1. Connections are made as shown in the circuit diagram.
2. Rated voltage of 110V is adjusted to get in voltmeter by adjusting the variac of the Auto Transformer which would be in zero before switching on the supply at the primary side.
3. The readings of voltmeter, ammeter and wattmeter are noted on the primary side.
4. A voltmeter is connected across the secondary and with the secondary supply off i.e switch S is kept open. The voltmeter reading is noted.
5. If the reading of voltmeter reads higher voltage, the terminals of any one of secondary coil is interchanged in order that voltmeter reads zero.
6. The secondary is now switched on and switch is closed with variac of auto transformer is zero.
7. After switching on the secondary the variac of transformer (Auto) is adjusted so that full load rated secondary current flows.
8. Then the readings of wattmeter, Ammeter and voltmeter are noted.
9. The Percentage Efficiency and percentage regulation are calculated and equivalent circuit is drawn.

EFFICIENCY

Efficiency of the transformer is defined as:

$$\eta = \frac{\text{output power}}{\text{input power}} \{23\}$$

Interms of losses,

$$\eta = \frac{\text{output power}}{\text{output power} + \text{iron losses} + \text{copper losses}} \{24\}$$

REGULATION

From Figure 4 it can be seen that if the input voltage is held constant, the voltage at the secondary terminals varies with load. Regulation is defined as the change in magnitude of secondary (terminal) voltage, when the load is thrown off with primary voltage held constant. Since, the change in secondary voltage depends only on the load current, the equivalent circuit is further simplified and is shown in Figure 9.

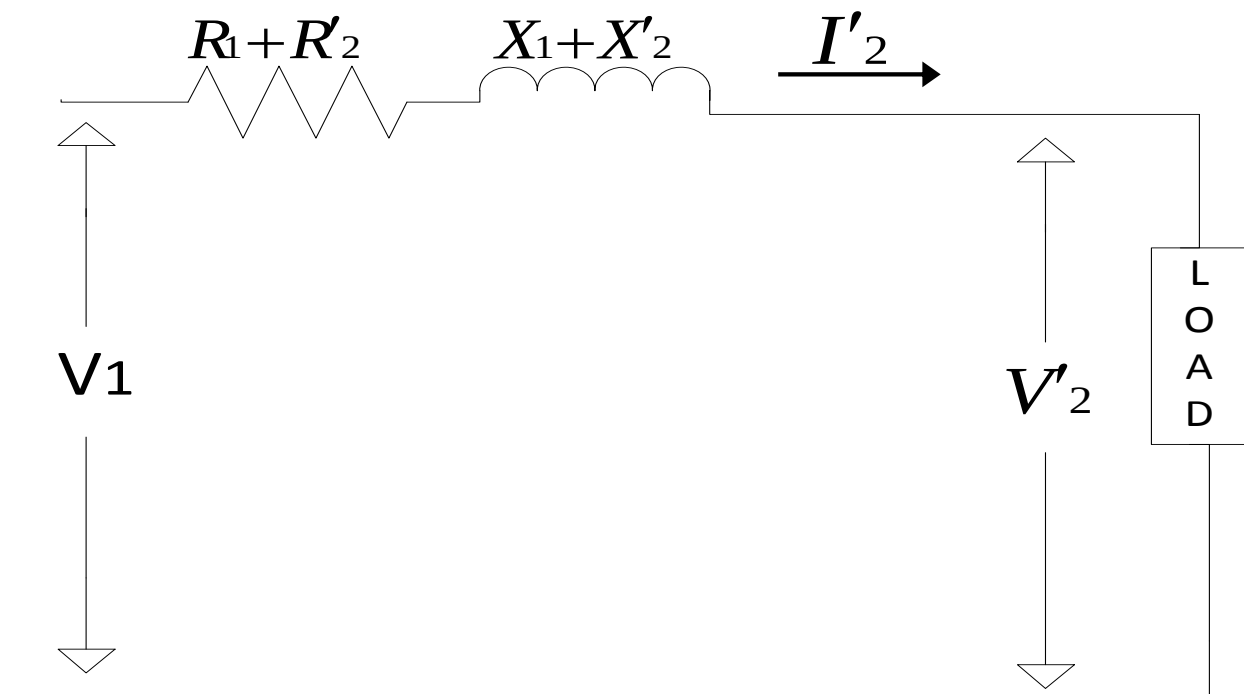


Figure 9: Equivalent circuit to determine regulation

The expression for regulation is given by

$$regulation = \frac{I_2 R_{eq} \cos \phi_o \pm I_2 X_{eq} \sin \phi_o}{V_2} \times 100 \quad \{25\}$$

MAGNETIC CIRCUIT

When magnetic force is applied to a material, the B and H fields which result are related according to the following expression:

$$B = \mu H \quad \{26\}$$

where $\mu = \mu_o \mu_r$ is permeability of the material.

$\mu_o = 4\pi \times 10^{-7}$ H/m and μ_r is the relative permeability of the material.

B = Flux density

H = Magnetic Field

The relative permeability for non-magnetic material is 1. The relative permeability of common transformer grade steel lamination, on the other hand, is in the order of thousands.

For a transformer having its primary winding excited with a current, I and its secondary winding open circuited (Refer to Figure 10), the relation between the peak magnetic field intensity, H_m and the exciting rms current, I is as follows:

$$H_m L = N_1 \sqrt{2} I_1 \quad \{27\}$$

The magnetic core of the transformer is assumed to have a uniform cross-section area and effective length of L . N_1 is the number of turns of the primary winding. Hence,

$$H_m \propto I \quad \{28\}$$

The value of electromotive force (e.m.f) induced in the secondary winding N_2 is

$$V_2 = 4.44 f A B_m N_2 \quad \{29\}$$

Where V_2 is secondary e.m.f. in r.m.s, f is the supply frequency and B_m is the maximum flux density in the transformer core.

Thus the B-H curve may be experimentally determined from simple voltage and current measurements.

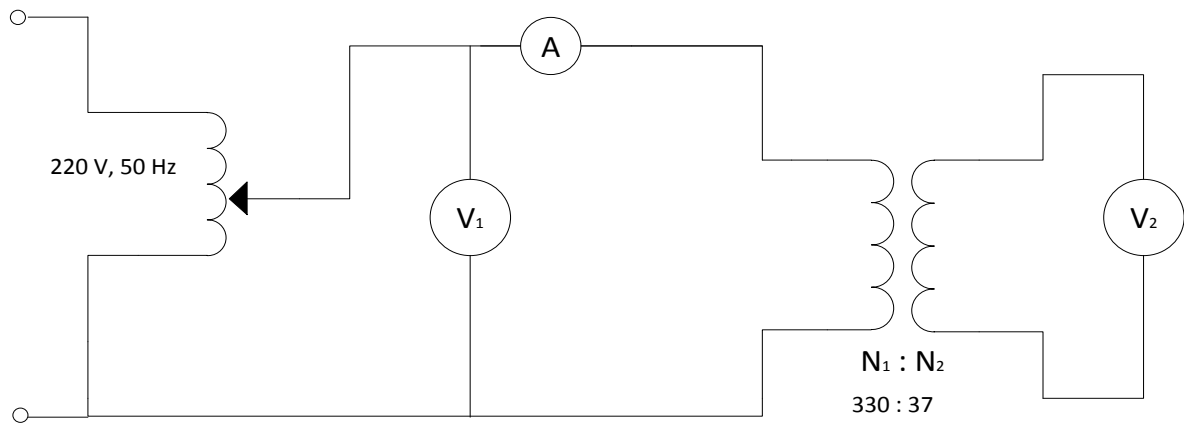


Figure 10: Circuit Diagram for B-H curve

PROCEDURE

1. The circuit as in Figure 10 is connected. The effective length, L and cross-section area, A are measured. $L=0.806$ m, $A=0.003276$ sq. m
2. The variac reading is set to zero and is switched on.
3. The variac is set to maximum voltage & corresponding current is noted ($I_{\max}=4$ A).
4. The variac is increased from 0 and the primary current, I_1 and the open-circuited secondary voltage, V_2 are measured.
5. The curves of B_m versus H_m are plotted.

APPARATUS REQUIRED

SL NO	APPARATUS	RANGE
1.	AC ammeter	(0-10) A
2.	AC Voltmeter	(0-300) V
3.	Autotransformer	6 A, (0-270)V
4.	Wattmeter	300 V, 1500 division
5.	Single Phase core type transformer	1 kVA, 220/24 V, 4.54/40.5 A, 330:37

PRE EXPERIMENT QUESTIONS:

1. What is the operating principle of a single phase transformer?
2. Draw & explain the phasor diagram of a single phase transformer for leading & lagging power factor.
3. Explain core & copper losses in a transformer & how these losses can be calculated with the help of Open circuit & short circuit test.
4. Draw the equivalent circuit of a transformer.

5. Why short circuit test is performed on high voltage side?
6. Why open circuit test is performed on low voltage side?
7. What will happen if primary of the transformer is connected to D.C. supply?
8. Why does the core of the transformer get saturated?
9. What is the purpose of conducting the Sumpner's test?
10. Why transformers are rated in KVA?

POST EXPERIMENT QUESTION

1. What are the ratings of the various instruments used during the experiment?
2. Why are iron losses constant at all loads in a transformer?
3. Why the B-H curve of the transformer does not depend upon the load?
4. Calculate the parameters of the transformer with the help of readings taken and draw phasor diagrams for OC and SC test using the calculated parameters.
5. How are the parameters referred to the HV or LV side? Explain with an example.
6. Why LPF wattmeter is used in OC test?
7. What are the advantages of Sumpner's test over OC & SC tests?
8. What is the difference between a single phase transformer & an autotransformer?
9. What changes do resistance & leakage reactance bring about in a practical loaded transformer?
10. What is the importance of referring the parameters of a transformer with respect to a particular side?

EXPERIMENTAL DATA

(For Open Circuit Test)

VOLTMETER READING (V _{oc}) Volts	AMMETER READING (I _{oc}) Ampere	WATTMETER READING (W _{oc}) Watts

(For Short Circuit Test)

VOLTMETER READING (V _{oc}) Volts	AMMETER READING (I _{oc}) Ampere	WATTMETER READING (W _{oc}) Watts

(For Sumpner Test)

Open circuit

VOLTMETER READING (V _{oc}) Volts	AMMETER READING (I _{oc}) Ampere	WATTMETER READING (W _{oc}) Watts

Short Circuit

VOLTMETER READING (V _{oc}) Volts	AMMETER READING (I _{oc}) Ampere	WATTMETER READING (W _{oc}) Watts

(For B-H Curve)

Primary Current, I ₁ (A)	Secondary Voltage, V ₂ (V)	Magnetic Force, $H_m = \frac{\sqrt{2}N_1 I_1}{L}$	Max. Flux Density, $B_m = \frac{V_2}{4.44fN_2 A}$